

SMART MULTI-SPECIALITY HOSPITAL NETWORK

NAT/PAT Configuration Summary — Per-Branch Internet Access with Inter-Branch Exclusion

Router	Site	Own Subnet	ACL Deny (No NAT)	ACL Permit (NAT)	NAT Interface
R1	HQ Edge	10.0.0.0/22	10.0.0.0/22 -> 10.0.0.0/8	10.0.0.0/22 -> any	Gi0/0 (overload)
R4	Diagnostic Center Edge	10.1.0.0/24	10.1.0.0/24 -> 10.0.0.0/8	10.1.0.0/24 -> any	Gi0/0 (overload)
R6	Emergency Clinic Edge	10.2.0.0/24	10.2.0.0/24 -> 10.0.0.0/8	10.2.0.0/24 -> any	Gi0/0 (overload)
R9	Telemedicine Edge	10.3.0.0/24	10.3.0.0/24 -> 10.0.0.0/8	10.3.0.0/24 -> any	Gi0/0 (overload)
R11	Maternity Hospital Edge	10.4.0.0/24	10.4.0.0/24 -> 10.0.0.0/8	10.4.0.0/24 -> any	Gi0/0 (overload)
R13	Children Care Hospital Edge	10.5.0.0/24	10.5.0.0/24 -> 10.0.0.0/8	10.5.0.0/24 -> any	Gi0/0 (overload)

Each edge router NATs only internet-bound traffic (destination outside 10.0.0.0/8) using PAT/overload on its own public IP. Traffic destined to another branch (still within 10.0.0.0/8) is explicitly denied in the ACL, so it is routed natively via EIGRP without translation, preventing inter-branch timeouts caused by double NAT.

NAT / PAT CONFIGURATION — CLI CODE (ALL EDGE ROUTERS)

Extended ACL excludes inter-branch (10.0.0.0/8) traffic from NAT overload

R1 — HQ Edge (10.0.0.0/22)

```
en
conf t

no ip nat inside source list 1 interface g0/0 overload
no access-list 1

ip access-list extended NAT_ACL
  deny ip 10.0.0.0 0.0.3.255 10.0.0.0 0.255.255.255
  permit ip 10.0.0.0 0.0.3.255 any
exit

int g0/1
  ip nat inside
exit

int g0/0
  ip nat outside
exit

ip nat inside source list NAT_ACL interface g0/0 overload
end
```

R4 — Diagnostic Center Edge (10.1.0.0/24)

```
en
conf t

ip access-list extended NAT_ACL
  deny ip 10.1.0.0 0.0.0.255 10.0.0.0 0.255.255.255
  permit ip 10.1.0.0 0.0.0.255 any
exit

int g0/1
  ip nat inside
exit
int g0/0
```

```
ip nat outside  
exit
```

```
ip nat inside source list NAT_ACL interface g0/0 overload  
end
```

R6 — Emergency Clinic Edge (10.2.0.0/24)

```
en
conf t

ip access-list extended NAT_ACL
  deny ip 10.2.0.0 0.0.0.255 10.0.0.0 0.255.255.255
  permit ip 10.2.0.0 0.0.0.255 any
exit

int g0/1
  ip nat inside
exit

int g0/0
  ip nat outside
exit

ip nat inside source list NAT_ACL interface g0/0 overload
end
```

R9 — Telemedicine Edge (10.3.0.0/24)

```
en
conf t

ip access-list extended NAT_ACL
  deny ip 10.3.0.0 0.0.0.255 10.0.0.0 0.255.255.255
  permit ip 10.3.0.0 0.0.0.255 any
exit

int g0/1
  ip nat inside
exit

int g0/0
  ip nat outside
exit

ip nat inside source list NAT_ACL interface g0/0 overload
end
```

R11 — Maternity Hospital Edge (10.4.0.0/24)

```
en
conf t

ip access-list extended NAT_ACL
  deny ip 10.4.0.0 0.0.0.255 10.0.0.0 0.255.255.255
  permit ip 10.4.0.0 0.0.0.255 any
exit

int g0/1
  ip nat inside
exit

int g0/0
  ip nat outside
exit

ip nat inside source list NAT_ACL interface g0/0 overload
end
```

R13 — Children Care Hospital Edge (10.5.0.0/24)

```
en
conf t

ip access-list extended NAT_ACL
  deny ip 10.5.0.0 0.0.0.255 10.0.0.0 0.255.255.255
  permit ip 10.5.0.0 0.0.0.255 any
exit

int g0/1
  ip nat inside
exit

int g0/0
  ip nat outside
exit

ip nat inside source list NAT_ACL interface g0/0 overload
end
```

EIGRP MISCONFIGURATION FIXES

Corrected network statements for routers with mismatched subnet assignments

Issue: Router 9 (Telemedicine Core) and Router 11 (Maternity Core) were originally configured with network statements copied from a different site (R12/R13 subnets), which did not match their actual interface IP addresses. This prevented EIGRP neighbor relationships from forming on those links. The fixes below remove the incorrect statements and apply the correct ones matching each router's real subnet.

Fix for Router 9 (Telemedicine Core — was misconfigured with R12/R13 subnets)

```
en
conf t

router eigrp 100
no network 200.1.1.16 0.0.0.3
no network 10.255.0.16 0.0.0.3
network 10.255.0.12 0.0.0.3
network 10.3.0.0 0.0.0.255
end
```

Fix for Router 11 (Maternity Core — was misconfigured with R13 subnets)

```
en
conf t

router eigrp 100
no network 200.1.1.20 0.0.0.3
no network 10.255.0.20 0.0.0.3
network 10.255.0.16 0.0.0.3
network 10.4.0.0 0.0.0.255
end
```

After applying each fix, verify with `show ip protocols` to confirm the correct networks are listed, and `show ip eigrp neighbors` to confirm the neighbor relationship forms.